

Practice Problems:
Laws and Simple Circuits
Introduction to Electric Circuits
Supplemental Instruction

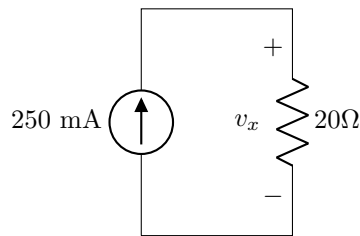
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February 19 2026

This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

1 Ohm's Law

Find v_x .

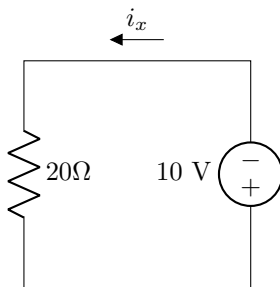


Solution

$$v_x = 250\text{mA} \cdot 20\Omega = 5 \text{ V}$$

2 Ohm's Law

Find i_x .



Solution

$$i_x = \frac{-10\text{V}}{20\Omega} = -500\text{mA}$$

3 Ohm's Law in Series

Find V_1 , V_2 , and V_3 .

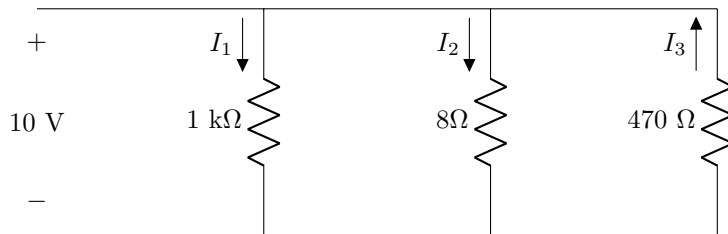


Solution

$$\begin{aligned} V_1 &= 50 \text{ mA} \cdot 100 \Omega = 5 \text{ V} \\ V_2 &= -50 \text{ mA} \cdot 2 \text{ k}\Omega = -100 \text{ V} \\ V_3 &= 50 \text{ mA} \cdot 7 \text{ M}\Omega = 350 \text{ kV} \end{aligned}$$

4 Ohm's Law in Parallel

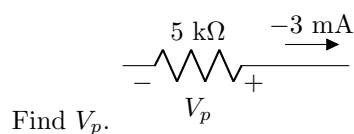
Find I_1 , I_2 , and I_3 .



Solution

$$\begin{aligned} I_1 &= \frac{10 \text{ V}}{1 \text{ k}\Omega} = 10 \text{ mA} \\ I_2 &= \frac{10 \text{ V}}{8 \Omega} = 1.25 \text{ A} \\ I_3 &= -\frac{10 \text{ V}}{470 \Omega} = -21.28 \text{ mA} \end{aligned}$$

5 Ohm's Law



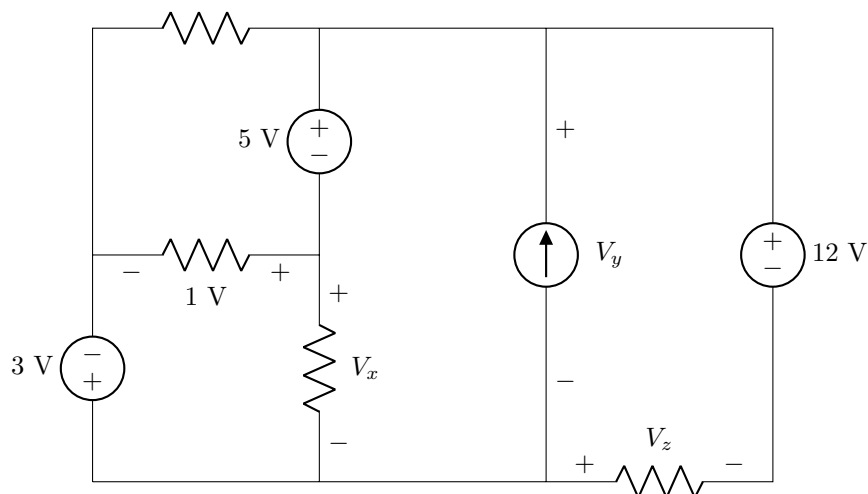
Find V_p .

Solution

$$V_p = -(-3 \text{ mA}) \cdot 5\text{k}\Omega = 15 \text{ V}$$

6 KVL

Find V_x , V_y , and V_z .



Solution

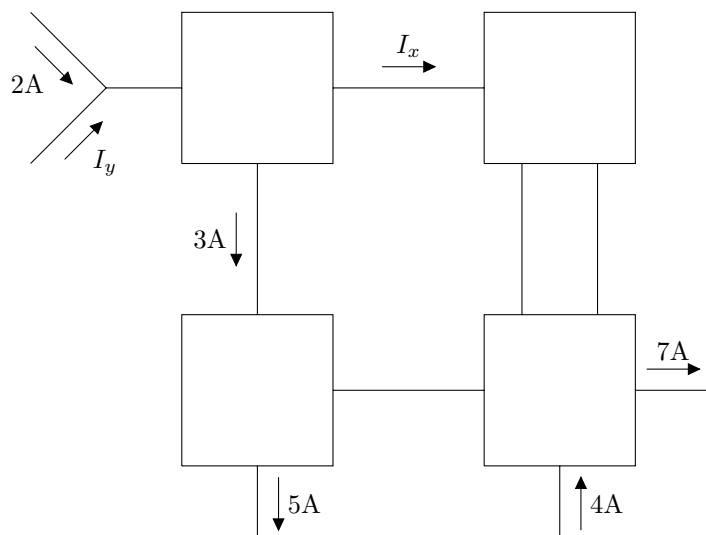
$$V_x = 1 - 3 = -2 \text{ V}$$

$$V_y = 5 + 1 - 3 = 3 \text{ V} \quad \text{or if we use } V_x, \quad V_y = 5 + (-2) = 3 \text{ V}$$

$$V_z = 3 - 1 - 5 + 12 = 9 \text{ V} \quad \text{or if we use } V_y, \quad V_z = -3 + 12 = 9 \text{ V}$$

7 KCL with Networks

Find (a) I_x , and (b) I_y . The boxes represent circuit networks.

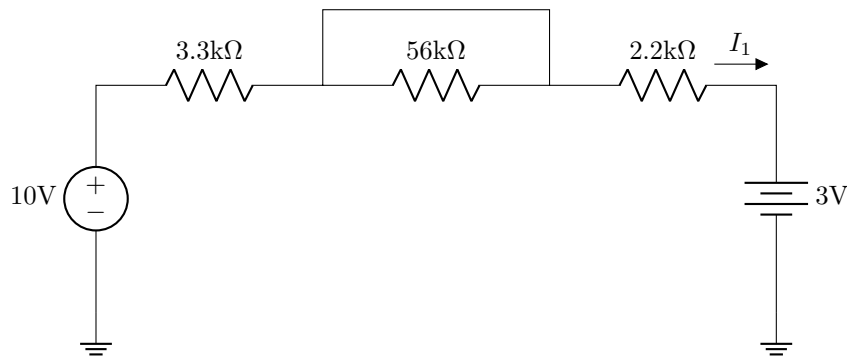


Solution

Summing the currents entering the network consisting of the bottom-right three boxes: $3\text{A} + I_x - 7\text{A} + 4\text{A} - 5\text{A} = 0 \Rightarrow I_x = 5\text{A}$ (a). Summing the currents entering the top-left box: $2\text{A} + I_y - I_x - 3\text{A} = 0 \Rightarrow I_y = 6\text{A}$ (b).

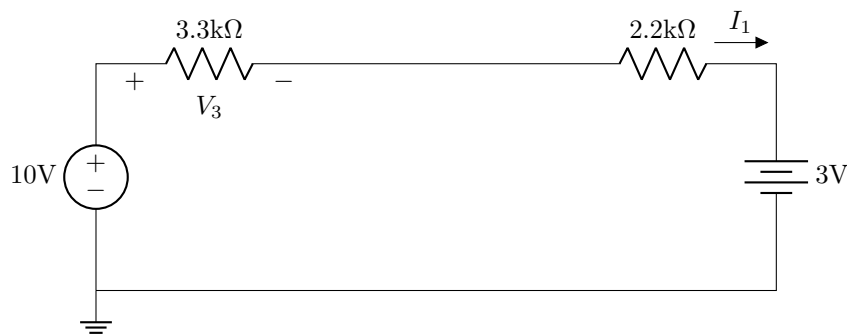
8 Finding current with KVL

Find I_1 , the power absorbed by the $3.3\text{k}\Omega$ resistor, and the power generated by the 10V source.



Solution

The circuit can be redrawn as:

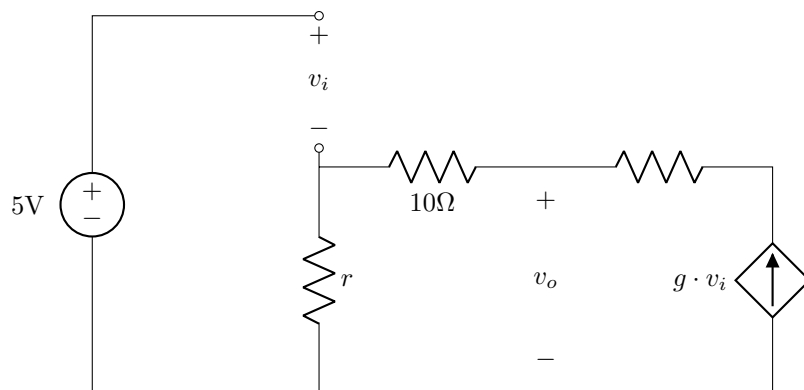


So, $I_1 = \frac{10\text{V} - 3\text{V}}{3.3\text{k}\Omega + 2.2\text{k}\Omega} = 1.27 \text{ mA}$.

The voltage on the $3.3\text{k}\Omega$ resistor is $V_3 = 1.27\text{mA} \cdot 3.3\text{k}\Omega = 4.191 \text{ V}$. Therefore the power absorbed is $P = iv = 1.27\text{mA} \cdot 4.191\text{V} = 5.32 \text{ mW}$. The power generated by the 10V source is $P = -iv = -1.27\text{mA} \cdot (-10\text{V}) = 12.7 \text{ mW}$.

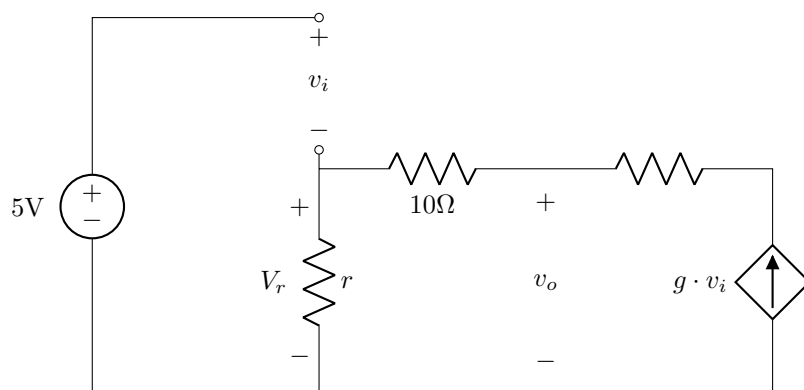
9 KVL Equations

Find v_o if $r = 2\Omega$ and $g = 5\text{S}$.



Solution

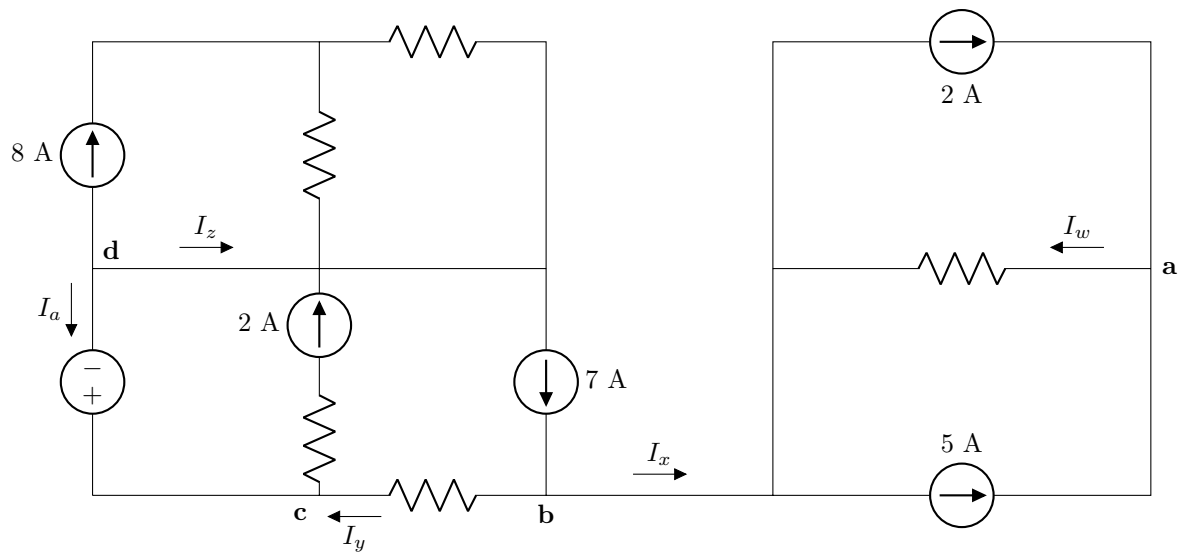
We can find the voltage across the resistor in two ways: using KVL, and using Ohm's Law. Let's first label this voltage:



Using KVL, $V_r = -v_i + 5\text{V}$. Using Ohm's Law, $V_r = g \cdot v_i \cdot r$. Equating the two, we have: $-v_i + 5\text{V} = 5\text{S} \cdot v_i \cdot 2\Omega$. Solving, $v_i = \frac{5}{11}\text{V}$. This means that the current $g \cdot v_i = \frac{25}{11}\text{A}$. Next, we can use KVL to find: $v_o = g \cdot v_i \cdot 10\Omega + V_r = \frac{25}{11} \cdot 10 + \frac{25}{11} \cdot 2 = 27.3\text{V}$.

10 KCL

Find I_w , I_x , I_y , and I_z .

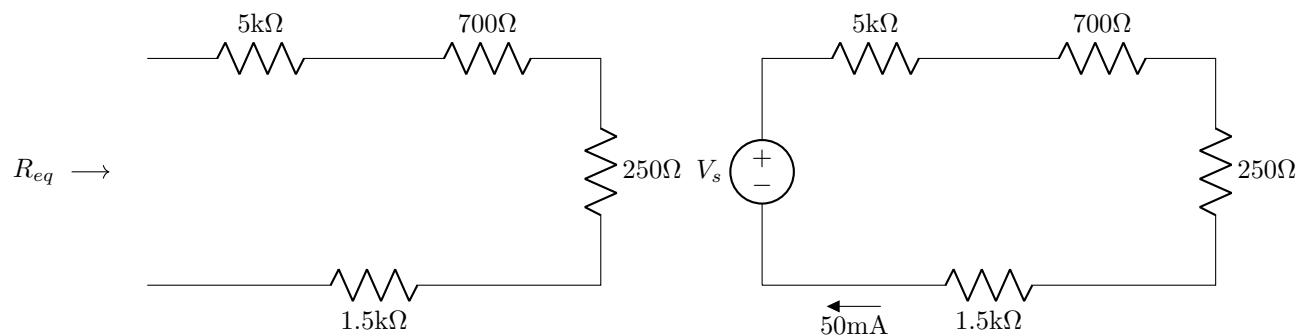


Solution

- (a) Summing currents entering at **a**: $2 + 5 - I_w = 0 \Rightarrow I_w = 2 + 5 = 7 \text{ A}$
 (b) Summing for either half of the circuit: $I_x = 0 \text{ A}$
 (c) Summing currents entering at **b**: $7 - I_x - I_y = 0 \Rightarrow I_y = 7 \text{ A}$
 (d) Summing currents entering at **c**: $I_y + I_a - 2 = 0 \Rightarrow I_a = 2 - 7 = -5 \text{ A}$
 then summing currents leaving at **d**: $I_a + I_z + 8 = 0 \Rightarrow I_z = -8 - (-5) = -3 \text{ A}$

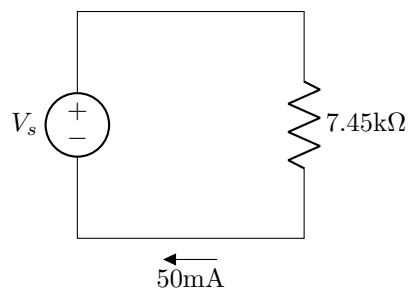
11 Series Resistor Combinations

Find R_{eq} and V_s .



Solution

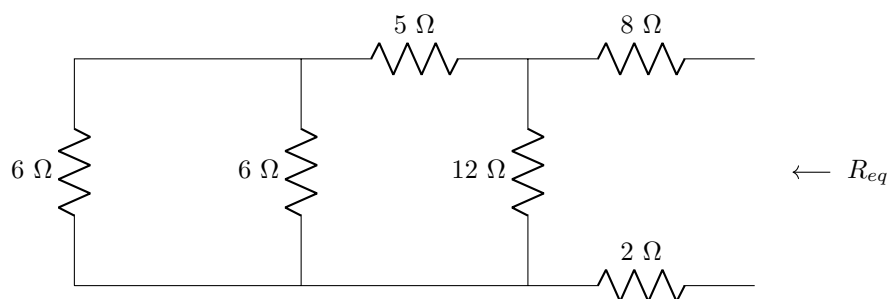
$R_{eq} = 5\text{k}\Omega + 700\Omega + 250\Omega + 1.5\text{k}\Omega = 7.45\text{k}\Omega$. We can draw the the equivalent circuit:



$$V_s = 50\text{mA} \cdot 7.45\text{k}\Omega = 372.5 \text{ V}$$

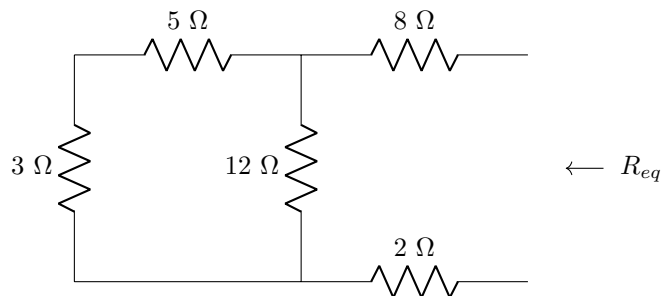
12 Resistor Combinations

Find R_{eq} .

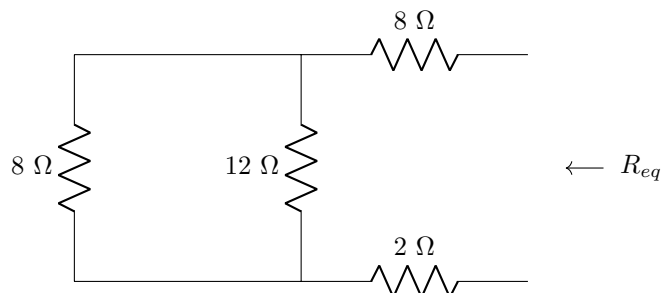


Solution

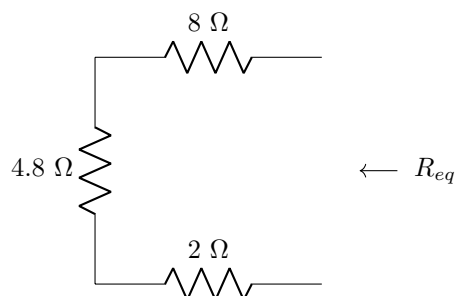
Combining the left two resistors in parallel: $\frac{1}{\frac{1}{6} + \frac{1}{6}} = 3$. We now have



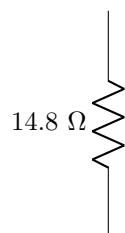
Combining the left two resistors in series, $5 + 3 = 8$. We now have



Combining the left two resistors in parallel, $\frac{1}{\frac{1}{8} + \frac{1}{12}} = 4.8$. We now have



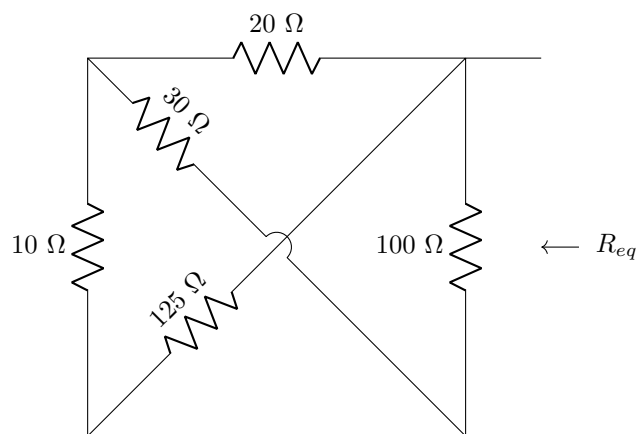
Combining the resistors in series, $8 + 4.8 + 2 = 14.8$.



$$R_{eq} = 14.8 \, \Omega$$

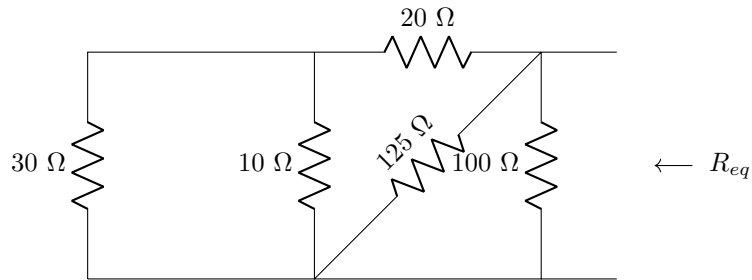
13 Resistor Combinations

Find R_{eq} .

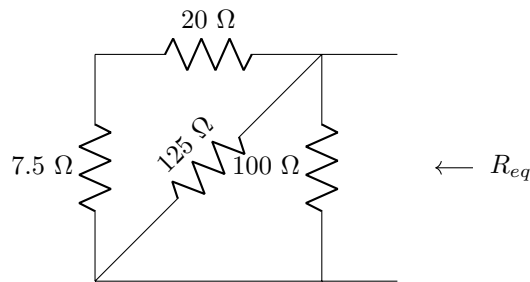


Solution

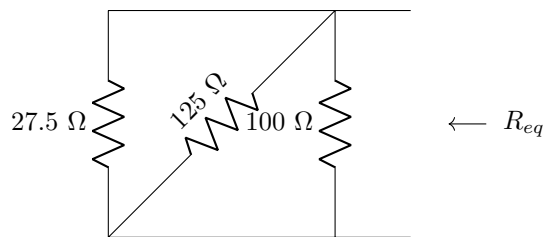
It will help to redraw the circuit by moving the $30\ \Omega$ resistor over to the left:



Combining the left two resistors in parallel, $\frac{1}{\frac{1}{30} + \frac{1}{10}} = 7.5$. We now have



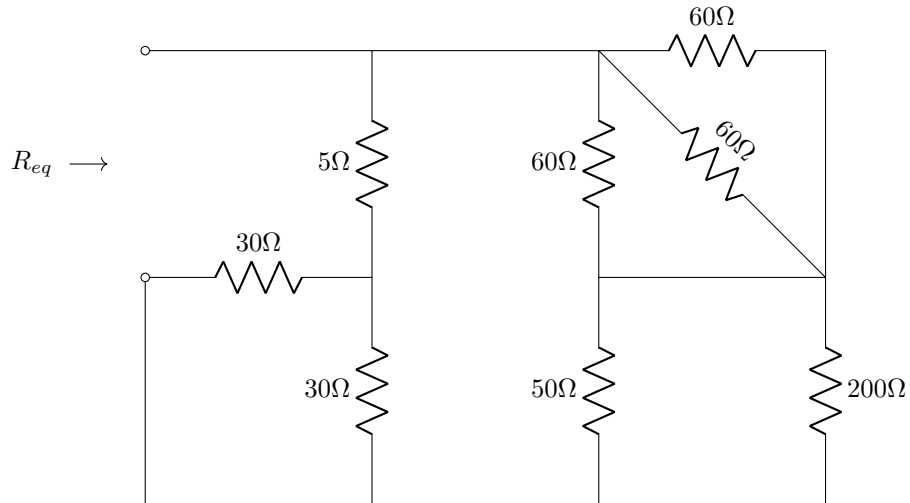
Combining the left two resistors in series, $7.5 + 20 = 27.5$. We now have



Combining the resistors in parallel, $\frac{1}{\frac{1}{27.5} + \frac{1}{125} + \frac{1}{100}} = 18.39$. $R_{eq} = 18.39\ \Omega$.

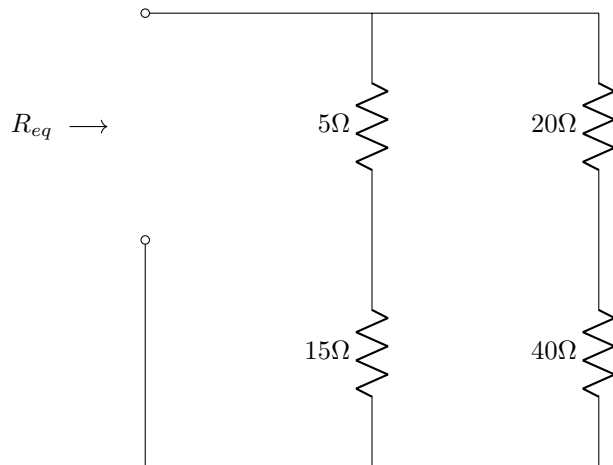
14 Resistor Combinations

Find R_{eq} .

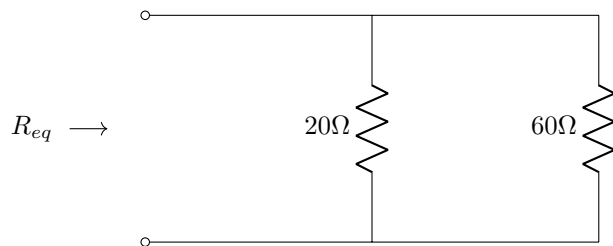


Solution

We can combine parallel resistors to obtain:



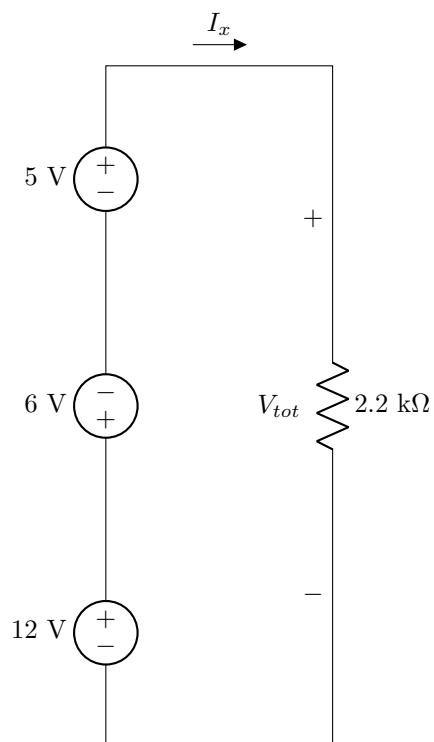
Then, combining series resistors,



Finally, combining parallel resistors, $R_{eq} = 15\Omega$.

15 Series Source Combinations

Find V_{tot} and I_x .



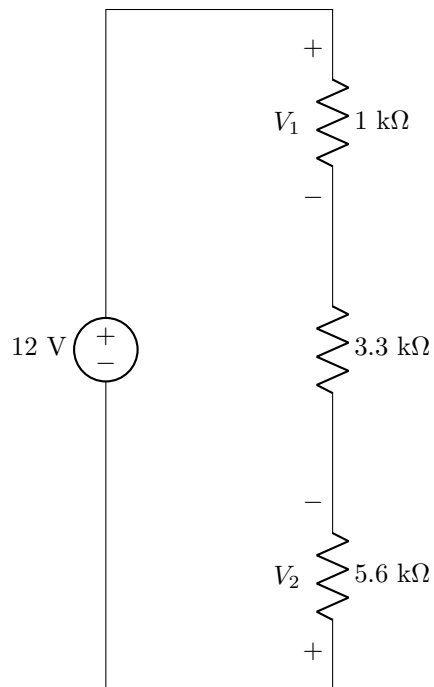
Solution

$$V_{tot} = 5 - 6 + 12 = 11 \text{ V}$$

$$I_x = \frac{V_{tot}}{2.2 \text{ k}\Omega} = 5 \text{ mA}$$

16 Voltage Division

Find V_1 and V_2 .



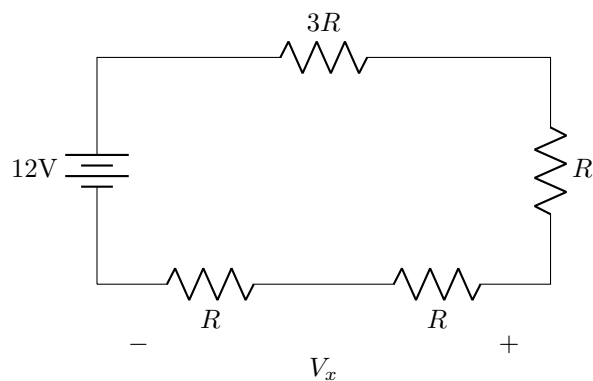
Solution

$$V_1 = 12\text{V} \cdot \frac{1\text{k}\Omega}{1\text{k}\Omega + 3.3\text{k}\Omega + 5.6\text{k}\Omega} = 1.212\text{ V}$$

$$V_2 = -12\text{V} \cdot \frac{5.6\text{k}\Omega}{1\text{k}\Omega + 3.3\text{k}\Omega + 5.6\text{k}\Omega} = -6.79\text{ V}$$

17 Voltage Division with unknown resistor values

Find V_x .

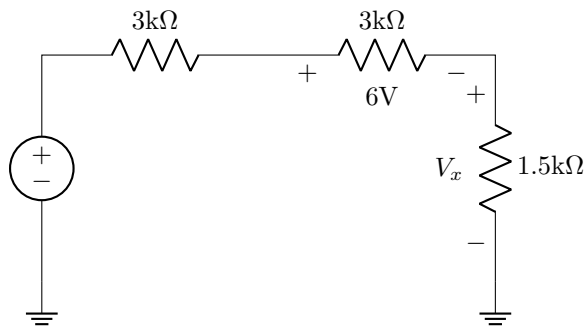


Solution

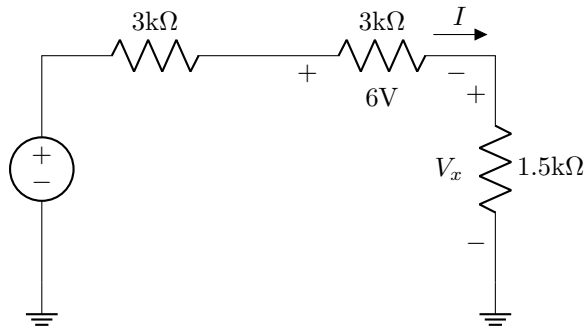
$$V_x = 12\text{V} \cdot \frac{R+R}{R+R+R+3R} = 12 \cdot \frac{2}{6} = 4\text{ V}$$

18 Voltage Division / Ohm's Law

Find V_x .

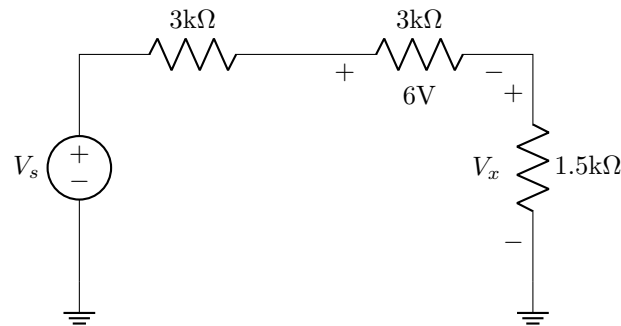


Solution (Method 1 - Ohm's Law)



Using Ohm's Law, $I = \frac{6\text{V}}{3\text{k}\Omega} = 2\text{mA}$. Then, $V_x = 2\text{mA} \cdot 1.5\text{k}\Omega = 3\text{ V}$.

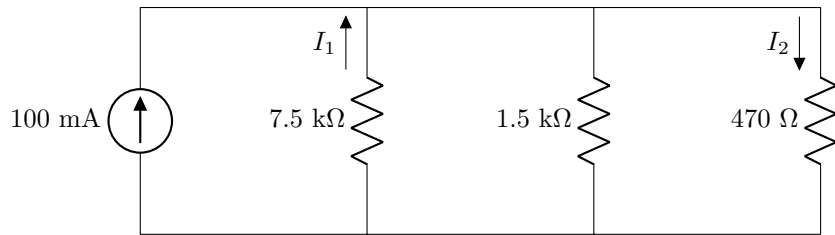
Solution (Method 2 - Voltage Division)



Using Voltage Division, $6\text{V} = V_s \cdot \frac{3\text{k}}{3\text{k}+3\text{k}+1.5\text{k}} \Rightarrow V_s = 15\text{V}$. Then, $V_x = 15\text{V} \cdot \frac{1.5\text{k}}{1.5\text{k}+3\text{k}+3\text{k}} = 3\text{ V}$.

19 Current Division

Find I_1 and I_2 .



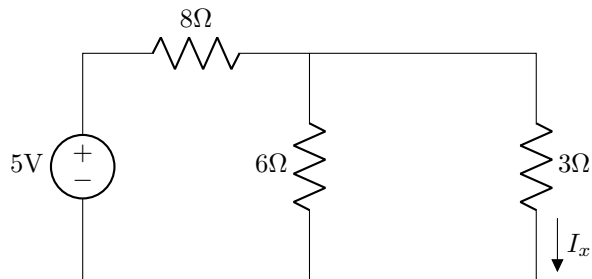
Solution

$$I_1 = -100\text{mA} \cdot \frac{\frac{1}{7.5\text{k}\Omega}}{\frac{1}{7.5\text{k}\Omega} + \frac{1}{1.5\text{k}\Omega} + \frac{1}{470\Omega}} = -4.55 \text{ mA}$$

$$I_2 = 100\text{mA} \cdot \frac{\frac{1}{470\Omega}}{\frac{1}{7.5\text{k}\Omega} + \frac{1}{1.5\text{k}\Omega} + \frac{1}{470\Omega}} = 72.67 \text{ mA}$$

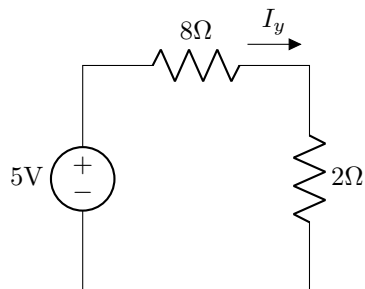
20 Voltage and Current Division with Ohm's Law

Find I_x .

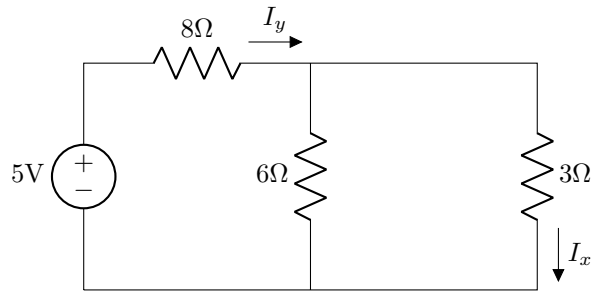


Solution (method 1)

Combining the resistors on the right: $\frac{1}{\frac{1}{6} + \frac{1}{3}} = 2\Omega$. Now we have



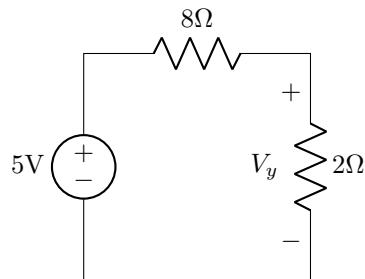
Combining resistors in series, $2+8 = 10\Omega$. Using ohm's law, we can find the total current $I_y = \frac{5\text{V}}{10\Omega} = 500\text{mA}$. We can now use current division:



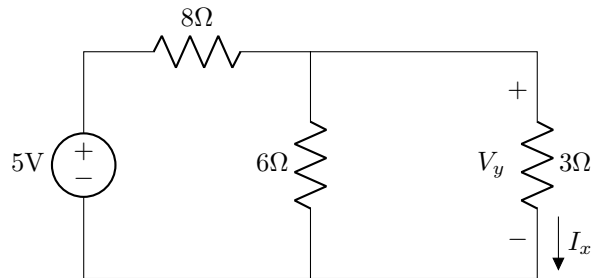
$$I_x = I_y \cdot \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{6}} = 500\text{mA} \cdot \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{6}} = 333.3\text{mA}$$

Solution (method 2)

Combining the resistors on the right:



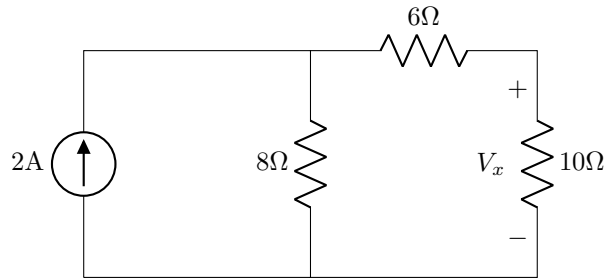
Using voltage division, $V_y = 5\text{V} \cdot \frac{2}{2+8} = 1\text{V}$. We can now use Ohm's law:



$$I_x = \frac{V_y}{3\Omega} = \frac{1\text{V}}{3\Omega} = 333.3\text{mA}$$

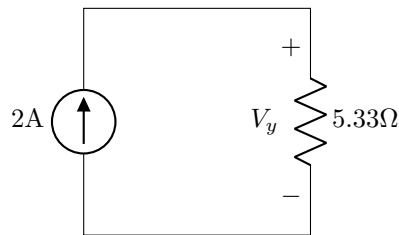
21 Voltage and Current Division with Ohm's Law

Find V_x .

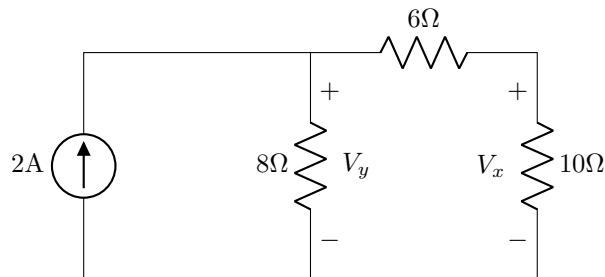


Solution

Combining resistors on the right, $6 + 10 = 16\Omega$. Then, combining resistors in parallel, $\frac{1}{\frac{1}{8} + \frac{1}{16}} = 5.33\Omega$. We now have



Using Ohm's Law, $V_y = 2A \cdot 5.33\Omega = 10.67V$. We can now use voltage division:

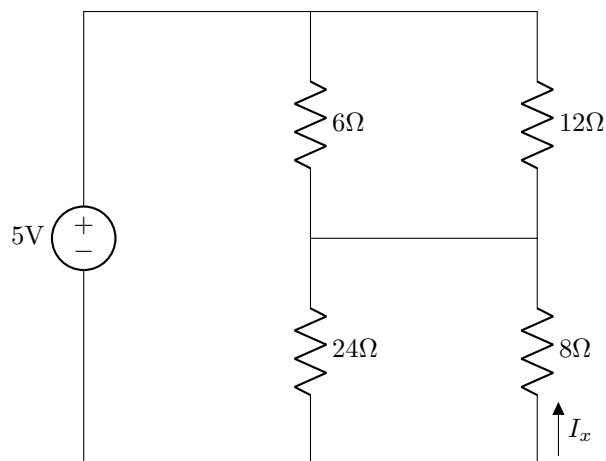


$$V_x = V_y \cdot \frac{10}{10 + 6} = 10.67V \cdot \frac{10}{10 + 6} = 6.67V$$

This problem could also be solved by combining the rightmost two resistors, using current division to find the current flowing through the rightmost branch, then using Ohm's Law to find the voltage on the 10Ω resistor.

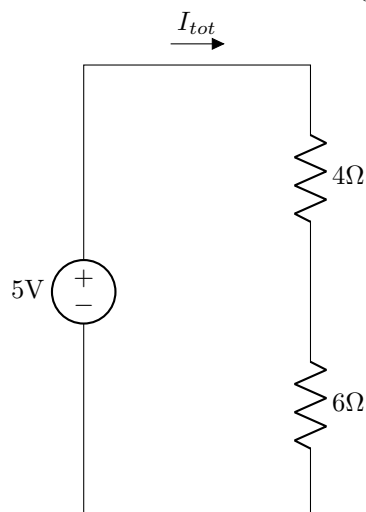
22 Current Division and Ohm's Law

Find I_x .



Solution

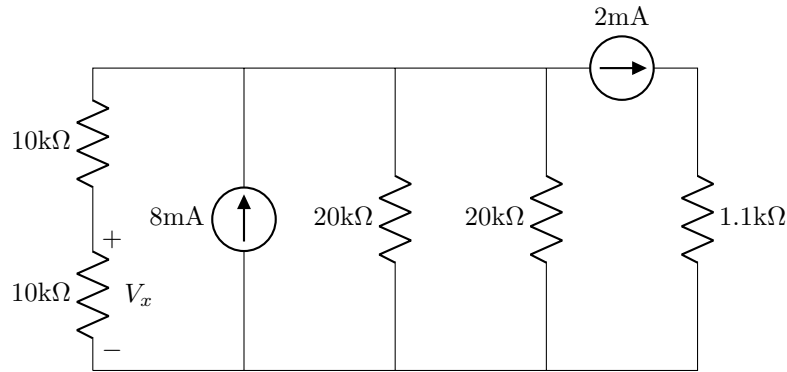
Combining parallel resistors, $\frac{1}{\frac{1}{6} + \frac{1}{12}} = 4\Omega$ (for the top), and $\frac{1}{\frac{1}{24} + \frac{1}{8}} = 6\Omega$ (for the bottom). We now have



Using Ohm's Law, $I_{tot} = \frac{5V}{(4+6)\Omega} = 500 \text{ mA}$. Using current division, $I_x = -500\text{mA} \cdot \frac{24}{24+8} = -375 \text{ mA}$.

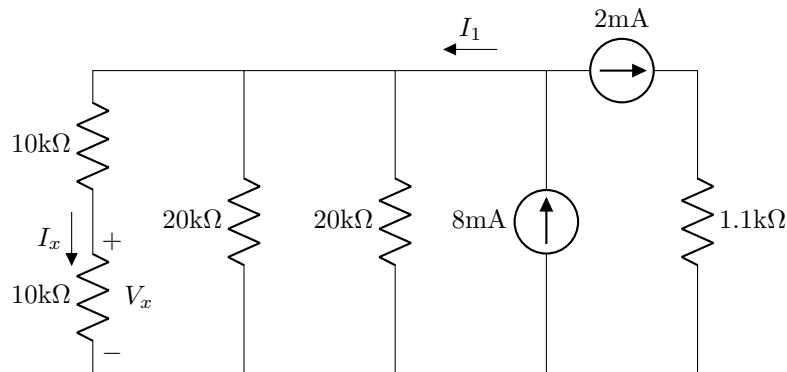
23 KCL, Current Division, Ohm's Law

Find V_x .



Solution

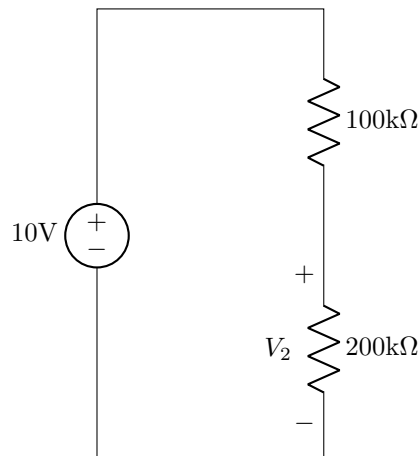
We can redraw the circuit as follows:



By KCL, $8\text{mA} = I_1 + 2\text{mA} \Rightarrow I_1 = 6\text{mA}$. Then, using current division, $I_x = \frac{6\text{mA}}{3} = 2\text{mA}$. Using Ohm's Law, $V_x = 2\text{mA} \cdot 10\text{k}\Omega = 20\text{ V}$

Loading Effect

A voltmeter with an internal resistance of $1\text{M}\Omega$ is connected across the $200\text{k}\Omega$ resistor. Find V_2 before and after the voltmeter is connected.

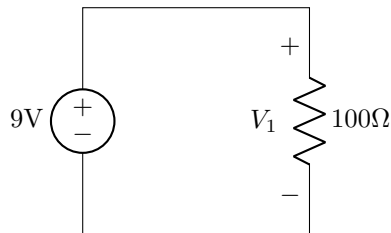


Solution

Before the voltmeter is connected, $V_2 = 10\text{V} \cdot \frac{200\text{k}\Omega}{100\text{k}\Omega + 200\text{k}\Omega} = 6.667\text{ V}$. After the voltmeter is connected, $V_2 = 10\text{V} \cdot \frac{200\text{k}\Omega \parallel 1\text{M}\Omega}{100\text{k}\Omega + (200\text{k}\Omega \parallel 1\text{M}\Omega)} = 6.25\text{ V}$.

24 Practical Voltage Source

Find V_1 when (a) the source has an internal resistance of 5Ω and (b) the source has an internal resistance of $10\text{m}\Omega$.

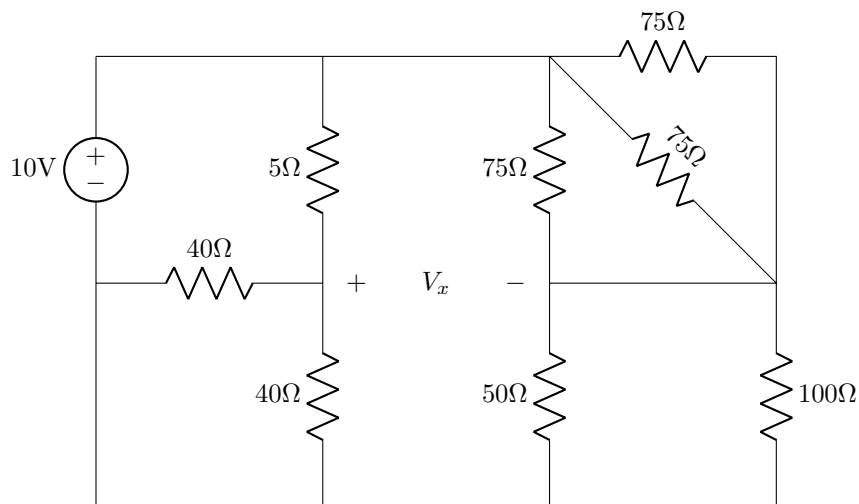


Solution

(a) $V_1 = 9\text{V} \cdot \frac{100\Omega}{100\Omega + 5\Omega} = 8.57\text{ V}$. (b) $V_1' = 9\text{V} \cdot \frac{100\Omega}{100\Omega + 10\text{m}\Omega} = 8.9991\text{ V}$.

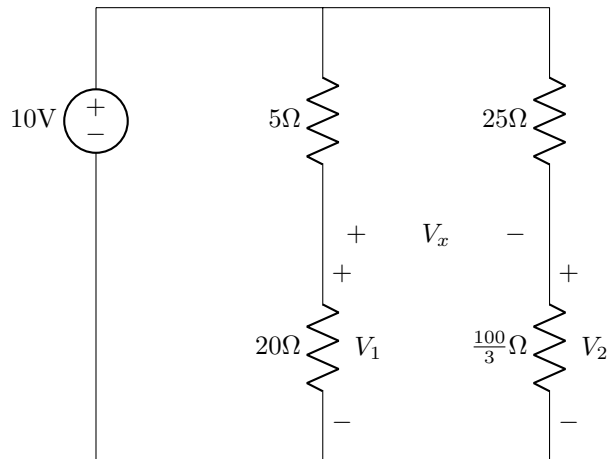
25 General Analysis

Find V_x .



Solution

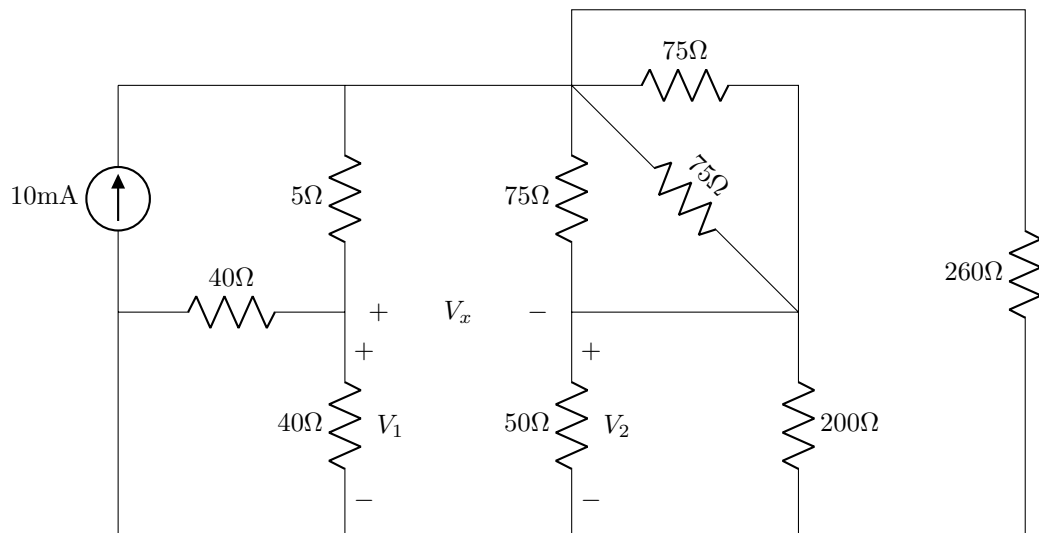
First we will combine all the parallel resistors:



We can find V_1 and V_2 using voltage division: $V_1 = 10 \cdot \frac{20}{20+5} = 8\text{V}$ and $V_2 = 10 \cdot \frac{\frac{100}{3}}{\frac{100}{3}+25} = \frac{40}{7}\text{V}$. Then, using KVL, $V_x = V_1 - V_2 = 8 - \frac{40}{7} = 2.29\text{ V}$.

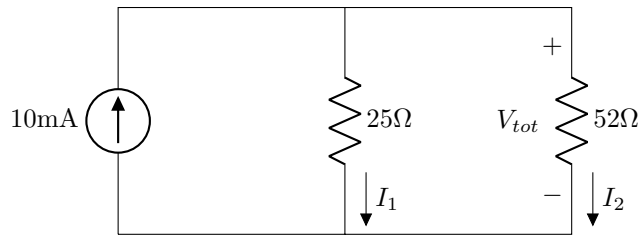
26 General Analysis

Find V_x .



Solution

We will solve this problem by finding first V_1 and V_2 . We can combine resistors to obtain:



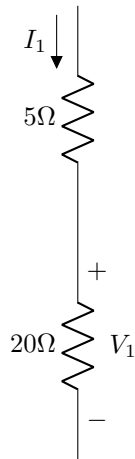
Using current division and KCL,

$$I_1 = 10\text{mA} \cdot \frac{\frac{1}{25}}{\frac{1}{25} + \frac{1}{52}} = 6.7532\text{mA}$$

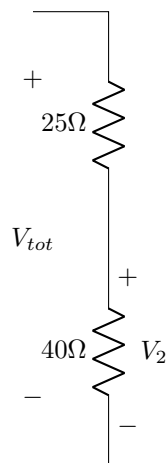
$$I_2 = 10\text{mA} - I_1 = 3.2467\text{mA}$$

We can also find the total voltage $V_{tot} = I_2 \cdot 52\Omega = 168.831\text{mV}$.

We can now use Ohm's Law to find V_1 , after combining the two 20Ω resistors:



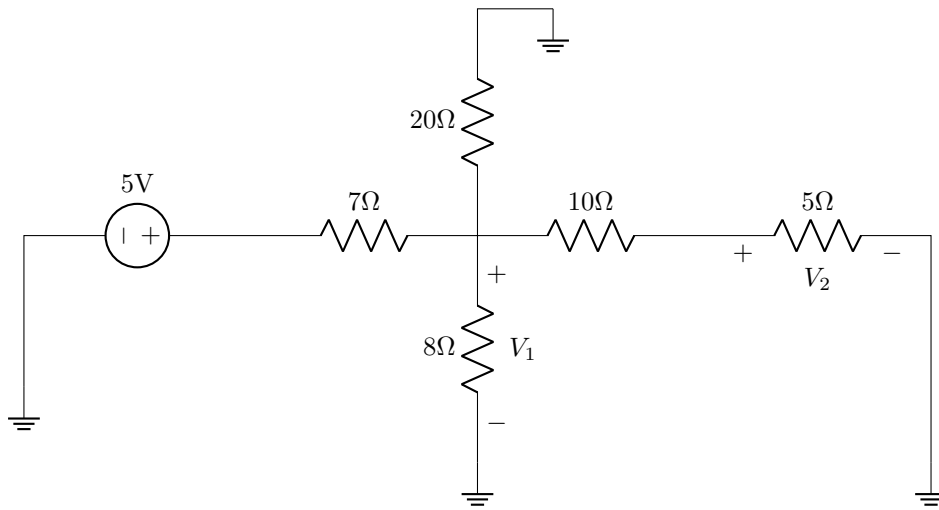
$V_1 = I_1 \cdot 20\Omega = 135.064\text{mV}$. Next, we can use voltage division to find V_3 :



$V_2 = V_{tot} \cdot \frac{40}{40+25} = 103.896\text{mV}$. Finally, using KVL, $V_x = V_1 - V_2 = 31.2\text{mV}$.

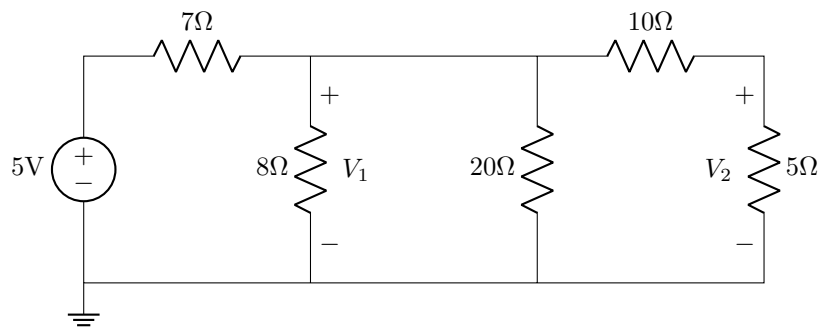
27 General Analysis

Find V_1 and V_2 .

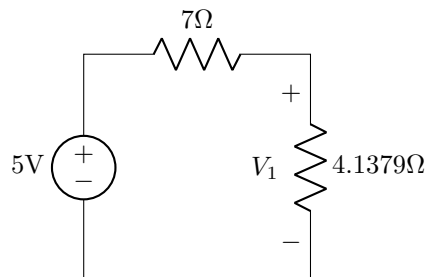


Solution

We can redraw the circuit as follows:



Combining resistors: $R_{eq} = \frac{1}{\frac{1}{8} + \frac{1}{20} + \frac{1}{10+5}} = 4.1379 \, \Omega$



Using voltage division, $V_1 = 5V \cdot \frac{4.1379}{4.1379+7} = 1.858 \, \text{V}$. Now, applying voltage division on the original simplified circuit: $V_2 = V_1 \cdot \frac{5}{5+10} = 1.858V \cdot \frac{5}{15} = 0.62V = 620 \, \text{mV}$.

References

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- M. Iordache. 2024. EEGR 2053 Electric circuits [PowerPoint slides]. Available: <https://mviordache.name/EEGR2053/index.html>
- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
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